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Summery Post

by Ali Alhammadi - Monday, 15 September 2025, 8:49 PM

Within the current discussion, it has been expounded how interaction can be made effective in multi-agent systems (MAS) by utilizing Agent Communication Languages (ACLs) (especially, KQML). As I have already said in my first posting, it has been observed that ACLs surpass message transmission regarding passing knowledge and intent. This makes them competent in open and heterogeneous situations such as supply-chain management and knowledge-sharing systems. They also have advantages such as standardization and scalability, which allow easy integration of more agents. I, too, however, found them wanting, especially in the computational cost of parsing and reasoning that restricts them be restricted to applications (in either runtime or performance sensitive) like robotics (Kuhail et al., 2023; Ma et al., 2024). ACLs also lack the flexibility and the ability to decide their own speed, as method invocation does in Python or Java and other such languages (Liu et al., 2025).

The peer feedback was more inclined to uphold this neutral argument and continue with the discussion fruitfully. My argument about scalability and interoperability was substantiated by other responses that indicated that ACLs were aimed or distributed applications. Others have characterized the inefficiency that complex logic ushered in, wherein others agreed that this makes ACLs less attractive when time dependence is sought. Among the similarities was the observation that the industry has a preference towards choosing lightweight, yet ACLs have a presence in the research.

Some peers also offered hybrid solutions, stating that combining the ideas of ACL and existing frameworks or machine-learned approaches, one could also reduce the number of semantics and yet perform computations. It implies the current-day conception that even though ACLs cannot be applied in their classical form all the time, it can still be designed in cooperation with the novel technologies (Kuhail et al., 2023; Liu et al., 2025). The feedback did not prove my simple arguments wrong, but merely complicated the analysis by demonstrating further directions and the real changes.

Overall, the discussion reveals the fact that there is consensus regarding the usefulness and difficulties of ACLs and peer responses concerning the opportunity to introduce a new act of their efficiency and real-life application.

References

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